# C H E M I S SI T R Y 17

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### Extracted Text

be in powder when shaken, it is a lign the operation bath fucceeded i but if it be in a cake, and duth not fall into powder on thinking the matras, it thews that your matter was not definitely youtted and dried in the iron pan du- ring the proprietary operation."

Mr Lemeri hath thewo, that excrement is not the only matter capable of producing this phofhorous with alumy but that, on the contrary, almost all animal and even vegetable matters are fit for that combinations; that though Mr Hongberg mixed alum in equal quantities only with the local master, it may be used in a much greater part portion, and you in certain cafes, will succeed the better; that, according to the nature of the fulbiances to be worked on, the quantity of that fish may be more or the increacity, and that whatever is indeed, more than the dofe requisite for each matter, ferives only to laffen the virtue of the photohorus, or even deflays it entirely that the degree of fire applied until the different according to the nature of those matters i and, kallly, that falls containing exactly the fame acid with that of alum, or the need of the folks far separated from his balls and reflects inno girls, too mustfer in the protein operation , which thews, fays Mr Lemeri, that many fulphureous matters may yet disbulated the excrement in this operation; but that there are no fails, or very few, if any, that will furced in the place of alum.

This phophony, make either by Mr Hongbergs or by Mr Lemers method, flines both by dry and by night. Besides counting light, it takes fire from after its is exposed to the air, and kindles all combultible matters with which it comes in control; and this without being rrabled for hexane.

Meff: Homberg and Lemeri have given the most pro-hable and the molt natural explanation of the course of the acordion and other phenomena of the pholphorus. What they lay amounts in floors or what follows.

Alom is known to be a neutral fats, confiling of the vintile aid and a deliberate earth. When the falls is calined with the fecal matter, or other fulbiances as bounding in oil, the volatile principles of their fulbiances, fact by the their plagey their files, and their oils, while in the fine manner so if they were difficiled; and then is nothing left in the matras, when thiefe principles are diffused, but a charred matter, like that which is found in recent wherein fach mixers have been decompelled by distillation.

This remainder therefore is nothing but a mixture of alum and churnals. Now, at the acid of the fils, which is the virtific, bath a greater affinity with the phologlinium than with any other fulbiance, it will quit its balls to unite with the phologlin of the coal, and he converted by that some into a fulphur . And that is the very cold of which we have certain proofs in the operation for pre- pairing this phorhons; the whom, after the volatile principle of the only matter we drawn off, the fire is increased, in order to combine closely together the fixed parts that remain in the matras, that sis, the alum and the shared matter, we perceive at the mouth of the matras a final fillon fulphourtsome, and a pungent finish of burning fulphar. Noy, when the operation is finih

ed, we find a real fulphur flicking in the neck of the ma-tras and, while the phofhorus is burning, it hasn plainly a frong fulphuceus finally. It is therefore cer-tain, that this phofhorus contains an adult fulphur; that is, a means ofgifted to take five with the spreift safe. But though fulphur be very inflammable, it never takes

fire of itself, without being either in contact with forms matter to actuality ignored, we still being expelled to a considerable degree of heat. Let us be then what may be the cause of its ascention, when it is a confitment part of the phofhorus.

We mention just now, that the acid of the alum quirs its balls, in order to form a fulphur by combining with the phoglition of the coal. This little we know to be an earth outside of being converted into lime, and that it is actually converted into quick-line by the calcination receifty to produce the phofhorus. We know that new made line, faith the property of writing with water to readily, that it thereby contradicts a very great degree of heat. Now when this phofhorus, which is partly continued of the basis of the alarm contained into quick-lines, is stopped to the salt, the time infanulty athetis the moilture of which the air is always full, and by this means, probably, grows to hot as to fire the fulphur which it is mixed. Perhaps also the acid of the alum is not totally changed into fulphur. Some parts thereof may be, only half-disfangged from its lands, and in that condition be capable of attracting flowingly the humidity of the air, of growing very hot likemically by up- bibing the moilture, and fo of contributing to the accen- fon of the phofhorus.

There is also more to think that all the phlogition of the charred matter is not employed in the production of fulphur in this, phofhorus, but that frame parts of it re-maning in the literal era can be coak.. The block colour of the unikalid phorhorus, and the red lipstick is emits white burning, sufficiently prove this, small air hunters.

Purs fome human urine into a glats lethic j fet it in a water bath, and difflr till there remain only about a function path of how you want, or able separate the urine in a pan feet in the _bal neam maria_ differ to reduced to the fame quantity. With this first nothing will ex- back but an initial phlegns, families houses like unipe. Xhi refridonium will, as the evaporation advances, because of a darker and darker roller, and at half acquire an al-mell black colors. Alongside this voldum with three its weight of fluids, and dilit is in a case for it in everbearing furnaces, with the usual precautions. At left there will come over a little more infidelp plegy like the fug-mer. When the matter is already, why a whole spirit will rise. After this spirit, white vapours will appear on interesting this free, a yellow oily liquor will come off, trickling down in veins i and together with the liquide a concrete volatile fate, which will likely to the fides of the receiver. At left there will come over a deep-colored feid old. In the resort there will remain a falling earthy colatum, which being illustrated, will yield some fen-fat.

### Matched Wikipedia Article

**Title:** Carbethopendecinium bromide
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## Source record

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be in powder when ſhaken, it is a ſign the operation hath ſucceeded: but if it be in a cake, and doth not fall into powder on ſhaking the matras, it ſhews that your matter was not ſufficiently roaſted and dried in the iron pan during the preparatory operation."

Mr Lemeri hath ſhewn, that excrement is not the only matter capable of producing this phoſphorus with alum; but that, on the contrary, almoſt all animal and even vegetable matters are fit for this combination; that though Mr Homberg mixed alum in equal quantities only with the fecal matter, it may be uſed in a much greater proportion; and, in certain caſes, will ſucceed the better; that, according to the nature of the ſubſtances to be worked on, the quantity of that ſalt may be more or leſs increaſed; and that whatever is added, more than the doſe requiſite for each matter, ſerves only to leſſen the virtue of the phoſphorus, or even deſtroys it entirely; that the degree of fire applied muſt be different according to the nature of thoſe matters; and, laſtly, that ſalts containing exactly the ſame acid with that of alum, or the acid of thoſe ſalts ſeparated from its baſis and reduced into ſpirit, do not anſwer in the preſent operation: which ſhews, ſays Mr Lemeri, that many ſulphureous matters may be ſubſtituted for excrement in this operation; but that there are no ſalts, or very few, if any, that will ſucceed in the place of alum.

This phoſphorus, made either by Mr Homberg's or by Mr Lemeri's method, ſhines both by day and by night. Beſides emitting light, it takes fire ſoon after it is expoſed to the air, and kindles all combuſtible matters with which it comes in contact; and this without being rubbed or heated.

Meſſ. Homberg and Lemeri have given the moſt probable and the moſt natural explanation of the cauſe of the accenſion and other phenomena of this phoſphorus. What they ſay amounts in ſhort to what follows.

Alum is known to be a neutral ſalt, conſiſting of the vitriolic acid and a calcareous earth. When this ſalt is calcined with the fecal matter, or other ſubſtances abounding in oil, the volatile principles of theſe ſubſtances, ſuch as their phlegm, their ſalts, and their oils, exhale in the ſame manner as if they were diſtilled; and there is nothing left in the matras, when thoſe principles are diſſipated, but a charred matter, like that which is found in retorts wherein ſuch mixts have been decompoſed by diſtillation.

This remainder therefore is nothing but a mixture of alum and charcoal. Now, as the acid of this ſalt, which is the vitriolic, hath a greater affinity with the phlogiſton than with any other ſubſtance, it will quit its baſis to unite with the phlogiſton of the coal, and be converted by that union into a ſulphur. And this is the very caſe, of which we have certain proofs in the operation for preparing this phoſphorus; for when, after the volatile principles of the oily matter are drawn off, the fire is increaſed, in order to combine cloſely together the fixed parts that remain in the matras, that is, the alum and the charred matter, we perceive at the mouth of the matras a ſmall blue ſulphureous flame, and a pungent ſmell of burning ſulphur. Nay, when the operation is finiſhed, we find a real ſulphur ſticking in the neck of the matras; and, while the phoſphorus is burning, it hath plainly a ſtrong ſulphureous ſmell. It is therefore certain, that this phoſphorus contains an actual ſulphur; that is, a matter diſpoſed to take fire with the greateſt eaſe. But though ſulphur be very inflammable, it never takes fire of itſelf, without being either in contact with ſome matter that is actually ignited, or elſe being expoſed to a conſiderable degree of heat. Let us ſee then what may be the cauſe of its accenſion, when it is a conſtituent part of this phoſphorus.

We mentioned juſt now, that the acid of the alum quits its baſis, in order to form a ſulphur by combining with the phlogiſton of the coal. This baſis we know to be an earth capable of being converted into lime; and that it is actually converted into quick-lime by the calcination neceſſary to produce the phoſphorus. We know that new made lime hath the property of uniting with water ſo readily, that it thereby contracts a very great degree of heat. Now when this phoſphorus, which is partly conſtituted of the baſis of the alum converted into quick-lime, is expoſed to the air, the lime inſtantly attracts the moiſture of which the air is always full, and by this means, probably, grows ſo hot as to fire the ſulphur with which it is mixed. Perhaps alſo the acid of the alum is not totally changed into ſulphur: ſome part thereof may be only half diſengaged from its baſis, and in that condition be capable of attracting ſtrongly the humidity of the air, of growing very hot likewiſe by imbibing the moiſture, and ſo of contributing to the accenſion of the phoſphorus.

There is alſo room to think that all the phlogiſton of the charred matter is not employed in the production of ſulphur in this phoſporus, but that ſome part of it remains in the ſtate of a true coal. The black colour of the unkindled phoſphorus, and the red ſparkles it emits while burning, ſufficiently prove this.

*Human Urine analyſed.*

Put ſome human urine into a glaſs alembic; ſet it in a water bath, and diſtil till there remain only about a fortieth part of what you put in; or elſe evaporate the urine in a pan ſet in the *balneum marie* till it be reduced to the ſame quantity. With this heat nothing will exhale but an inſipid phlegm, ſmelling however like urine. The reſiduum will, as the evaporation advances, become of a darker and darker ruſſet, and at laſt acquire an almoſt black colour. Mingle this reſiduum with thrice its weight of ſand, and diſtil it in a retort ſet in a reverberating furnace, with the uſual precautions. At firſt there will come over a little more inſipid phlegm like the former. When the matter is almoſt dry, a volatile ſpirit will riſe. After this ſpirit, white vapours will appear on increaſing the fire; a yellow oily liquor will come off, trickling down in veins; and together with this liquor a concrete volatile ſalt, which will ſtick to the ſides of the receiver. At laſt there will come over a deep-coloured fetid oil. In the retort there will remain a ſaline earthy reſiduum, which being lixiviated will yield ſome ſea-ſalt.

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Figure 1: *
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